

Name:
Z number:

Total Points: 26

COP 3410-001: Data Strc Algrthm Anlys Python

FLORIDA ATLANTIC UNIVERSITY

Binary Trees and Traversals

Worksheet

NOTE:

- Each question carries 1 point.
- Submit the completed worksheet in the form of a PDF.
- The file name must be *YourName_ Binary Trees and Traversals Worksheet.pdf*

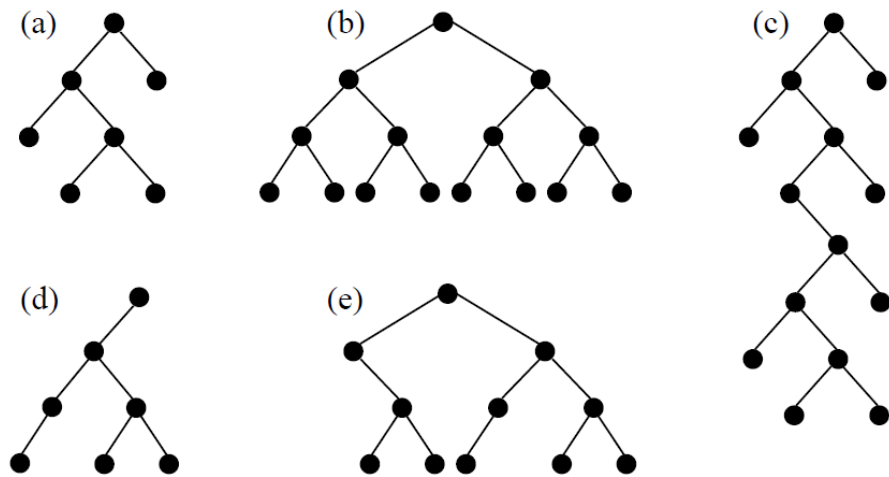
- 1- Given a binary tree of size 76, what is the minimum number of levels it can contain? What is the maximum number of levels? (1 points)

The minimum number of levels is 7

- 2- What is the maximum number of nodes possible in a binary tree with 5 levels? (1 point)

The maximum number of levels is 76

3- Given the following binary trees:



(a) Indicate all the structure properties that apply to each tree: *full*, *perfect*, *complete* (1 points).

full

(b) Determine the size of each tree (1 points).

Full, perfect, complete

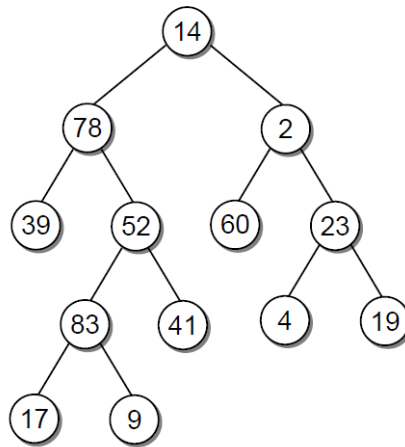
(c) Determine the height of each tree (1 points).

full

(d) Determine the width of each tree (1 points).

none

4- Consider the following binary tree:



(a) Show the order the nodes will be visited in a:

i. preorder traversal (1 point)

14, 78, 39, 52, 83, 17, 9, 41, 2, 60, 23, 4, 19

ii. inorder traversal (1 point)

39, 78, 17, 83, 9, 52, 41, 14, 60, 2, 4, 23, 19

iii. postorder traversal (1 point)

39, 17, 9, 83, 41, 52, 78, 60, 4, 19, 23, 2, 14

iv. breadth-first traversal (1 point)

14, 78, 2, 39, 52, 60, 23, 83, 41, 4,
19, 17, 9

(b) Identify all the leaf nodes (1 point).

39, 17, 9, 41, 60, 4, 19

(c) Identify all the interior nodes (1 point).

14, 78, 2, 52, 23, 83

(d) List all the nodes on level 4 (1 point).

17, 9

(e) List all the nodes in the path to each of the following nodes (1 points):

i. 83 14->78->52->83

ii. 39 14->78->39

(f) Consider node 52 and list the node's (1 points):

i. descendants 83, 41, 17, 9

ii. ancestors 14, 78

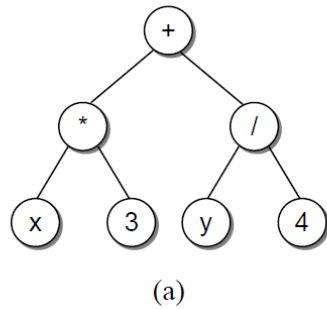
iii. Siblings 39

(g) Identify the depth of each of the following nodes (1 points):

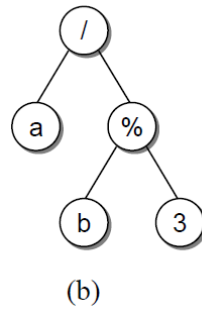
i. 78 depth of 1

ii. 41 depth of 3

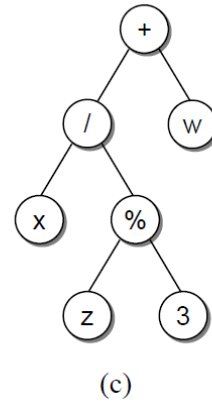
- 5- Determine the arithmetic expression represented by each of the following expression trees (3 points):



$(x*3)+(y/4)$



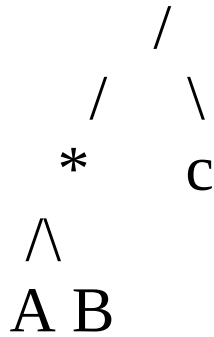
$a/(b\%3)$



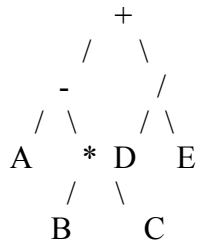
$(x/(z\%3))+w$

6- Build the expression tree for each of the following arithmetic expressions (3 points):

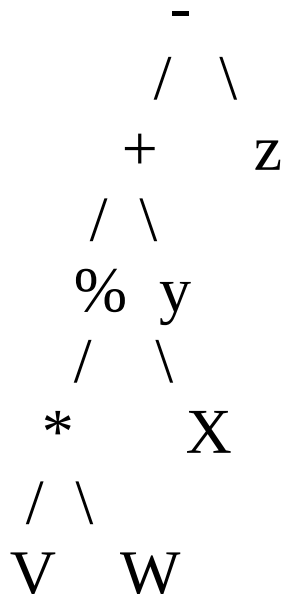
(a) $(A * B) / C$



(b) $A - (B * C) + D / E$



(c) $V * W \% X + Y - Z$



7- Consider the following set of values and use them to build a heap by adding one value at a time in the order listed:

30 63 2 89 16 24 19 52 27 9 4 45

(a) min-heap (1 point)

[2,4,19,27,9,30,24,89,52,63,16,45]

(b) max-heap (1 point)

[89,63,45,52,16,24,19,30,27,9,4,2]

- 8- Prove or show that the worst-case time of the extraction operation on a heap implemented as an array is $O(\log n)$. (1 point)

Reason 1: removing the top item and replacing it with the last item in the array takes a couple of quick array lookups, which takes a constant amount of time regardless of how big the heap is

Reason 2: the tree is very short because every level of heap holds twice as many items as the levels above

Reason 3: bubbling down only visit one path. In the worst case, that replacement item at the top is in the wrong spot and has to bubble down to the very bottom.

- 9- Prove or show that the insertion and extraction operations on a heap implemented as a Python list is $O(n)$ in the worst case. Also show that each operation has an amortized cost of $O(\log n)$. (1 point)

Worst-Case: Python's list runs out of empty memory space and has to pause and copy all n items to a new location taking $O(n)$ time

Amortized: full list copies happen so rarely that their cost averages out to nearly nothing. About $O(1)$ per operation. Everyday speeds depend on bubbling the item up or down the heap's tree which takes $O(\log n)$ time

Source: "Data Structures and Algorithm Analysis Using Python", R. Sedgwick, Wiley, 2011